

Assignment 6

1. Explain Store-and-Forward Packet Switching.
2. What services are provided by the Network Layer to the Transport Layer?
3. Explain IPv4 Addressing.
4. Explain the Implementation of Connection-less Service
5. Explain the Implementation of Connection-Oriented Service.
6. Explain IPv6 Addressing.
7. Differentiate between Virtual Circuit and Datagram Networks.
8. What is Classful Addressing?
9. Explain Subnetting and Supernetting.
10. Explain Classless Addressing (CIDR).
11. A block of addresses is granted to a company. One of the addresses in the block is **192.168.10.66/27**. Find the **First and last address** in the block.
12. Given the IP address **172.16.0.0/22**, calculate:
 - Number of subnets
 - Number of hosts per subnet
 - Valid host range for the first subnet
13. Given the IP address **10.0.0.0/27**, calculate:
 - Number of subnets
 - Number of hosts per subnet
 - Subnet mask in dotted decimal form